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Facility-level energy-driven water footprint and scarcity implications of Chinese data centers: a bottom-up analysis and scenario-based projection

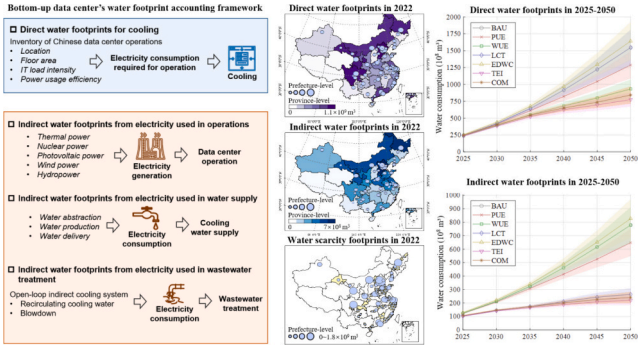
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HIGHLIGHTS

- A bottom-up framework estimates facility-level water footprints of Chinese data centers.
- 21 scenarios project future water footprints and regional water scarcity implications.
- 72 % of China's computing capacity is located in severe water-scarce regions by 2022.
- Simultaneous improvements in PUE, WUE, and grid decarbonization cut WFs by 51.5–58.3 %.

GRAPHICAL ABSTRACT



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ABSTRACT

Data centers have rapidly expanded to meet increasing computing demands. As energy-intensive infrastructures, data centers consume significant amounts of water to meet cooling requirements and support operations. Here, we present a bottom-up framework to estimate the facility-level water footprints (WFs) of data centers in China and project future pathways. First, we compile an inventory of Chinese data center operations and calculate both direct and indirect water consumption. Then, by mapping water scarcity levels, we assess the dependence of the data centers on limited water supplies. Finally, 21 scenarios are developed to model WFs trends and water scarcity implications from 2025 to 2050. The results show that in 2022, China's data centers consumed 15.7 billion m^3 of water, accounting for 2.7 % of the nation's total. Notably, 72 % of the computing capacity is located in severe water-scarce regions. By 2050, simultaneous improvements in power usage efficiency, water usage efficiency, and grid decarbonization could reduce national WFs by 51.5–58.3 % compared with the baseness as usual scenario, alleviating the increased water consumption driven by the Eastern Data and Western Computing Project in water-scarce regions such as Gansu and Ningxia Provinces. This study fills a gap in understanding the water consumption of Chinese data centers, highlighting the urgent need to incorporate water resource limitations into planning for sustainable digital infrastructure.

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Nomenclature

Acronyms

IT	Information technology
EDWC	Eastern Data and Western Computing Project
PUE	Power usage efficiency
WUE	Water usage efficiency
WF	Water footprint
WSF	Water scarcity footprint
WC	Water consumption
WSI	Water stress index
GDP	Gross domestic product
BAU	Baseness as usual

Notations

ITLoad	IT load intensity
FA	Floor area of the data center
T	Operational time
EC	Electricity consumption
WCR	Water cooling requirement per unit of electricity consumption
ELEC1	Electricity consumed for the data center operation
ELEC2	Electricity consumed by water treatment plants for cooling water supply
ELEC3	Electricity consumed by wastewater treatment plants for wastewater treatment
MIX	The structure of electricity generation mix or water withdrawal mix
WI	Water use intensity
f	Electricity consumption per unit of water supply process
β	Rate of water evaporation from the cooling system
K	The ratio of dissolved solids in blowdown water to intake water
WW	Water withdrawal
WA	Water availability
c	Adjustment factor of water stress index

Subscript

op	Operation
op_IT	Operation of IT equipment
d	Data center type
p	Province
s	Power generation technology
N	Number
elec	Electricity
wa	Water
abst	Water abstraction
prod	Water production
deli	Water delivery
A	The average design power per rack
i	Data center code
*	Prediction value

1. Introduction

The rapid expansion of the information technology (IT) industry has driven explosive growth in the demand for supporting infrastructure. By 2024, the global IT market is projected to reach a global value of 344 billion USD [1]. Data centers serve as the information backbone of the increasingly digitalized world, housing IT equipment, such as servers, storage devices, and network hardware, alongside systems to meet their power and cooling demands [2]. With over 11,000 data centers operating worldwide [3], their expansion is increasingly linked to the widespread adoption of artificial intelligence, machine learning, and other data-intensive technologies [4] [5].

As essential infrastructure in the digital era, data centers are also significant consumers of electricity and water [6]. The continuous operation of servers and cooling systems contributes to approximately 1–2 % of global electricity use [7,8]. Their operations also lead to both direct and indirect water consumption. Data centers consume water directly for cooling purposes, which helps dissipate the immense heat produced during operation. Additionally, data centers consume water indirectly through the water requirements for electricity generation [9]. Electricity is used to support data center operations, supply treated water to data centers, and process wastewater discharged from data centers [10]. The global power sector is currently the second largest consumer of freshwater resources after agriculture [11]. The strong interconnection between energy and water systems has led to growing concerns about the environmental impacts of data centers [12–14].

China, which accounts for approximately 33 % of global computing capacity [15], is experiencing particularly rapid growth in its data center sector, with projections indicating a growth rate of 11.9 % from 2025 to 2030 [16]. However, this growth is spatially uneven: approximately 65 % of data centers are concentrated in the more economically developed eastern provinces, many of which face significant energy constraints [17,18]. To address these environmental impacts, the Chinese government launched the Eastern Data and Western Computing Project (EDWC) in 2022. This initiative aims to relocate data storage and computational resources from the eastern regions to the western regions. New data centers have been established in areas such as Gansu and Inner Mongolia to alleviate energy pressures in the east [19]. However, many of these new centers are located in water-scarce regions. China is one of the most water-scarce countries, with per capita water resources accounting for only a quarter of the global average [20]. These challenges underscore the urgent need for a spatially explicit analysis of water use across China's data centers.

The environmental impact of data centers has received academic coverage. Various studies and reports have analysed the energy consumption of data centers [21,22]. Within the global IT industry, data centers are the largest contributors to energy consumption and carbon footprint, accounting for up to 45 % [8,23]. While energy efficiency has long been emphasized, scholars have highlighted the need to consider water use in sustainable data center design [24]. However, analyses of data center's water consumption remains limited, with only rough global estimates available [25]. A key reason is the lack of transparency, as IT companies are not required to disclose water usage data [13,14,26]. This lack of data is particularly acute in China, where bottom-up information on data center locations, IT equipment types, and operational characteristics is not readily available. As a result, quantifying water consumption across Chinese data centers remains extremely challenging. Without the spatially detailed data, policymakers cannot effectively target areas for water-saving interventions or make informed decisions about where to site new data centers. Addressing this data gap is crucial for enabling more efficient resource allocation and advancing water-conscious digital infrastructure, particularly in water-stressed regions like western China.

In the assessment of water consumption, the water footprint is an effective metric for evaluating water use from a bottom-up perspective. The water footprint calculation process must not only account for the direct water consumption resulting from onsite operations but also consider the indirect water use associated with electricity consumption for operations, water supply, and wastewater treatment. For the latter, the electricity structure and generation technologies in the region where the data center is located play critical roles in influencing indirect water consumption [27]. This step highlights the importance of spatial

analysis in data center water resource management. The incorporation of the water footprint into the national regulatory framework for data center water use is crucial for promoting more sustainable water resource management.

In response to the water consumption challenges posed by data centers, improving power usage efficiency (PUE) and water usage efficiency (WUE) has become crucial [23]. Energy efficiency measures, such as increasing the penetration of renewable energy [28,29] and establishing centralized layouts [30], can promote the decoupling of service demand from electricity usage, thereby reducing the water consumption associated with electricity use in data centers. The Chinese government has announced specific targets for improving the PUE in various cities and has proposed national standards for the WUE [31,32]. Additionally, the ongoing transition in China's electricity structure, driven by carbon neutrality goals, indirectly affects the energy and water use of data centers [22,33]. The EDWC Project may exacerbate the uneven regional water scarcity. The negative impacts of the exponential growth of computational power in the future may not be fully compensated for by

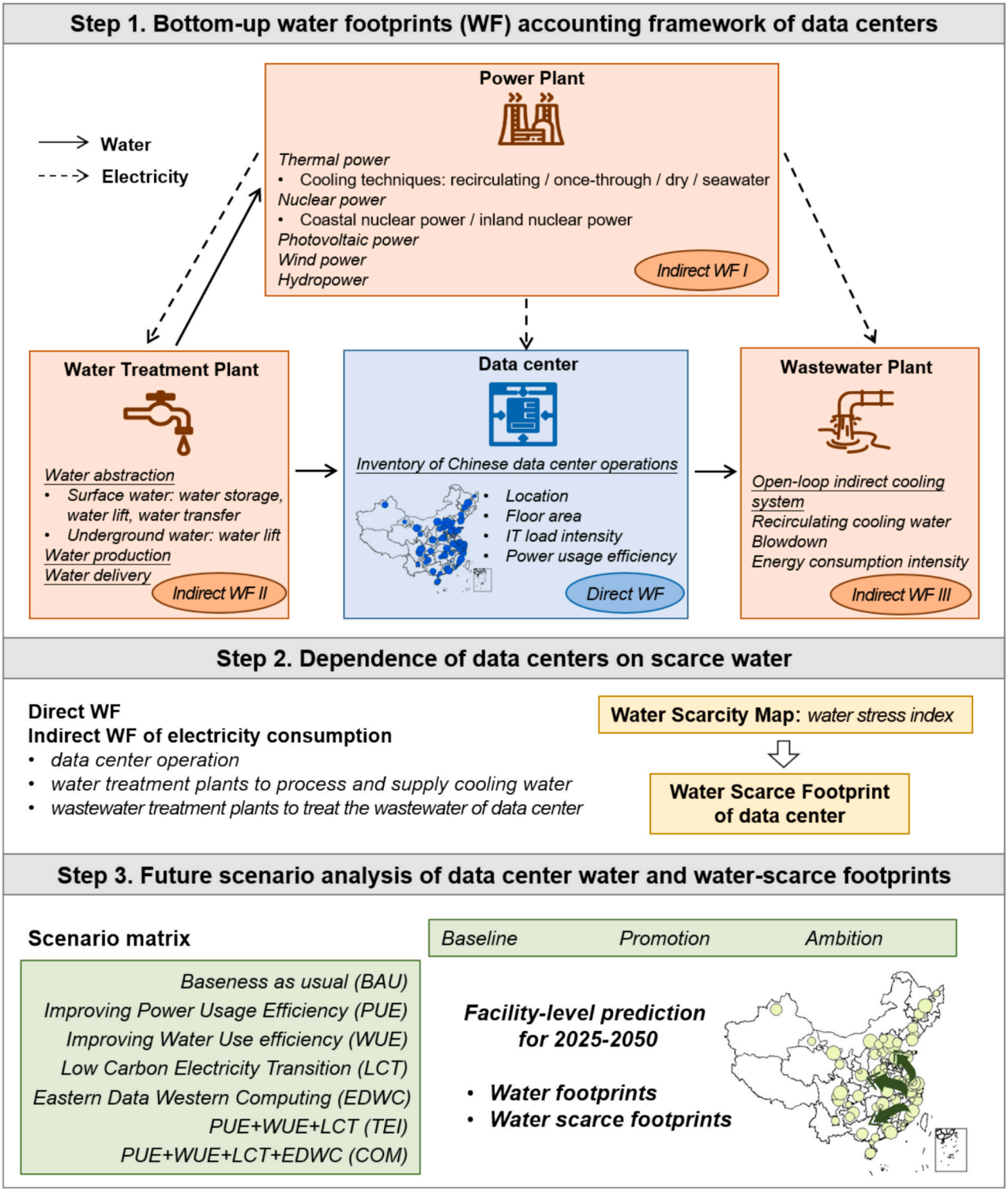


Fig. 1. Technical framework.

the efficiency gains from technological innovations in data centers [34]. Under the complex influence of these factors, the evolution of the water footprint of data centers in China remains uncertain.

To fill these gaps, we established a spatially detailed record of the operations of data centers in China and developed a bottom-up water footprint (WF) accounting framework. Based on this framework, we estimate the current and future facility-level water footprints (WFs) of data centers. We explore the following questions: (i) What are the direct and indirect water footprints of Chinese data centers, and how are they spatially distributed? (ii) How does this water consumption affect regional water scarcity? (iii) How does the strategic placement of future data centers in China affect WFs? (iv) Considering the evolving levels of future water scarcity, how do existing strategies exacerbate or alleviate regional water stress?

2. Methods

We collect the facility-level operational data of Chinese data centers. Based on these data, we develop a bottom-up framework to assess both the current and future WFs and water scarcity implications of data centers, as illustrated in Fig. 1. This framework consists of three main components: first, we construct the WFs inventory for each data center, including both direct and indirect water footprints; second, we integrate the Chinese prefecture-level water scarcity data to establish a water scarcity footprint (WSF) inventory for these data centers; finally, taking into account government policies for energy and water conservation in data centers and the EDWC strategy, we develop 21 scenarios to analyze the future evolution of WFs and their impact on regional water scarcity.

2.1. Water footprints estimation

The WF of data centers is defined as follows. (1) Direct WF: the onsite water consumption directly used for the operation of data centers, such as cooling, that is, WC^{op}. (2) Indirect WF: the offsite water consumption resulting from the electricity used by the data center, consisting of WC^{elec1}, WC^{elec2}, and WC^{elec3}.

$$WF = WC^{op} + WC^{elec1} + WC^{elec2} + WC^{elec3} \tag{1}$$

(1) Direct WF.

The cooling water consumption of data centers is estimated based on the heat generation capacity during computing processes [35], which is related to the electricity consumption during operations [36]. The electricity consumption consists of two parts: the electricity used by IT equipment and the electricity used by other auxiliary equipment, such as cooling and power distribution.

$$EC^{op_IT} = ITLoad_d \times FA \times T \tag{2}$$

$$ELEC1 = EC^{op_IT} \times PUE_d \tag{3}$$

where EC^{op_IT} represents the IT equipment electricity consumption (in MWh) during the operational phase of data centers and $ELEC1$ represents the total electricity consumption (in MWh) of all equipment during the operational phase. According to the International Data Corporation classification system (Table S2), data centers are classified based on the floor area of their server rooms, with d representing the type. $ITLoad_d$ denotes the IT load intensity values (ITs in watts/m^2) corresponding to data center type d. FA indicates the floor area of the data center's server room (m^2), and T represents the operational time. The PUE is defined as the ratio of total energy consumption to the energy consumption of IT equipment. A PUE greater than 1 and closer to 1 indicates higher energy efficiency, indicating less energy consumption by non-IT equipment.

The cooling water consumption during the operational phase of the data center (WC^{op}, m^3) is calculated by multiplying the IT equipment electricity consumption by the water cooling requirement per unit of electricity consumption (WCR). The value of WCR is 1.8 m^3/MWh [9,37].

$$WC^{op} = EC^{op_IT} \times WCR \tag{4}$$

(2) Indirect WF.

The indirect WF of data centers includes the water consumption associated with the following electricity usages: (i) the electricity consumed during the operation of the data center ($ELEC1$), (ii) the electricity used by water treatment plants to process and supply cooling water to the data center ($ELEC2$), and (iii) the electricity used by wastewater treatment plants to treat the wastewater produced by the data center ($ELEC3$).

For (i), the offsite water consumption (WC^{elec1}) resulting from $ELEC1$ is related to the local power grid mix and the power generation technologies.

$$WC^{elec1} = \sum_{s=1}^{N} ELEC1 \times MIX_p^{elec} \times WI_{s,p} \tag{5}$$

For a data center, MIX_p^{elec} is the electricity structure of province p, which varies significantly across provinces in China [38] (Table S3). The water use intensity $WI_{s,p}$ represents the water used per unit of electricity generated by technology s in province p (m^3/MWh).

The main types of electricity generation in China include thermal power, nuclear power, wind power, photovoltaic power, and hydropower. Among these, thermal power, primarily coal-fired power, contributes the most to freshwater consumption in the sector. The WI of thermal power depends mainly on the cooling type adopted in coal-fired power plants, which can be categorized into four types: recirculating cooling, once-through cooling, dry cooling, and seawater cooling (Table S4, S5). The WI of nuclear power depends on its geographical location, i.e., whether it is coastal or inland [39]. Additionally, photovoltaic power stations require periodic cleaning of photovoltaic panels, resulting in a certain amount of water consumption, with the WI is approximately 0.02 m^3/MWh [40]. The water consumption of wind power and hydropower is negligible. Therefore, $N = 7$ represents the set of coastal nuclear power, inland nuclear power, photovoltaic power generation and four cooling technologies for coal-fired power.

For (ii), the $ELEC2$ includes the electricity consumed for water abstraction, water production, and water delivery processes. The electricity consumption for water abstraction varies depending on the type of water source. For surface water, the electricity consumption processes include water storage, water lifting, and water transfer. For underground water, the electricity consumption process mainly involves water lifting. The electricity consumption for other stages can be considered negligible.

$$ELEC2 = WC^{op} \times MIX_p^{wa} \times f_p^{abst} + WC^{op} \times f_p^{prod} + WC^{op} \times f_p^{deli} \tag{6}$$

The water withdrawal mix MIX_p^{wa} of province p represents the proportion of surface water and underground water. f_p^{abst} denotes the electricity consumption per unit of water abstraction (kWh/m^3). Similarly, f_p^{prod} represents the electricity consumption per unit of water production (kWh/m^3), and f_p^{deli} denotes the electricity consumption per unit of water delivery (kWh/m^3). The data are provided in Table S6. Therefore, the water consumption resulting from $ELEC2$ can be calculated as Eq. (7).

$$WC^{elec2} = \sum_{s=1}^{N} ELEC2 \times MIX_p^{elec} \times WI_{s,p} \tag{7}$$

For (iii), the cooling system of the data center utilizes an open-loop indirect cooling process. To prevent the accumulation of dissolved solids, minerals, and other impurities in the water, blowdown is necessary during operation. This discharged water, referred to as blowdown water, is determined by the system's evaporation rate (β) and the concentration ratio (K). The corresponding electricity consumption $ELEC3$ can be obtained as Eq.(8), where f_p^{treat} is the electricity consumption per

unit of wastewater treatment. Furthermore, the indirect water consumption during the power generation process (WC^{elec3}) can be determined based on Eq.(9).

$$ELEC3 = WC^{op} \times \beta \times \frac{1}{K-1} \times f_p^{treat} \tag{8}$$

$$WC^{elec3} = \sum_{s=1}^{N} ELEC3 \times MIX_p^{elec} \times WI_{s,p} \tag{9}$$

2.2. Water scarce footprints estimation

The water scarce footprint (WSF), as defined by ISO 14046 [41], measures the impact of consumptive water use on the availability of freshwater in a river basin. Compared with the water footprint, the water scarcity footprint can more directly reflect the impact of data center water usage on local water scarcity. Referring to the method proposed by Pfister et al., we first calculate the water stress index (*WSI*), which is defined as the ratio of the sum of industrial water use, agricultural water use, and domestic water use (collectively termed water withdrawal, *WW*) to the water availability (WA) in the region [42,43].

$$WSI = \frac{WW}{WA} \tag{10}$$

$$WSL = \frac{1}{1 + e^{-c \bullet \mathrm{WSI}} \times \left(\frac{1}{0.01} - 1\right)} \tag{11}$$

$$WSF = WF^{*}WSL \tag{12}$$

The value of *WSI* is converted to [0,1] using Eq. (11): The parameter *c* is used to adjust the sensitivity of *WSL* to the *WSI*, thereby making the distinction between moderate and high stress more pronounced. The four water scarcity levels—low, moderate, severe, and extreme—correspond to *WSI* boundaries of 0.2, 0.4, and 1.0, respectively. By setting parameter *c*, the corresponding *WSL* values are 0.1, 0.5, and 0.9. The objective is to allow for a consistent comparison of water scarcity levels across different regions and to enhance the distinction between moderate and severe stress levels.

2.3. Water footprint prediction

The demand for computing power reflects the level of economic development in the digital era. The annual growth rate of the national computing scale is set to 2.5 times the annual GDP growth rate. This proportional relationship is derived from the China Academy of Information and Communications Technology's statistics, which indicate a positive correlation between China's computing power scale and economic development level [15]. We referred to the GDP prediction data under SSP scenarios [44]. Considering the uncertainty of this forecast, we set different growth rates for the Baseline, Promotion, and Ambition scenario sets (Section 2.4), which correspond to the SSP1, SSP2, and SSP5 development pathways, respectively.

Considering the future layout of computing power, i.e., the implementation of the East–West Computing Resource Transfer strategy at the provincial level, we estimate the province-level power consumption of IT equipment during the data center operation phase. As shown in Eqs. (13)–(14),

$$EC^{op_IT^{*}}{}_p = N_p \times A_p \times ITLoad_p \times T \tag{13}$$

$$ELEC1^{*}{}_p = EC^{op_IT^{*}}{}_p \times PUE_p \tag{14}$$

where $EC^{op_IT^{*}}$ represents the future power consumption of the IT equipment during the data center operation phase (in MWh). N_p is the number of racks in province *p*, which is used to reflect the scale of computing power. A_p is the average design power per rack in province *p*. The remaining variables have the same meanings as in the previous formulas.

We project facility-level operational electricity consumption by narrowing down the projected province-level $ELEC1^{*}{}_p$ to the facility level according to the current share of electricity consumption of the plants over the total provincial electricity consumption.

$$ELEC1^{*}{}_i = ELEC1^{*}{}_p \times \frac{ELEC1_i}{ELEC1_p} \tag{15}$$

As shown in Eq. (15), $ELEC1^{*}{}_i$ denotes the predicted electricity consumption of all equipment in data center *i* during the operational phase. After that, based on Eqs. (4)–(9), the facility-level water footprints of data centers can be predicted with specific parameter settings under the scenarios.

2.4. Adaptive water scarcity level for future WSF estimation

An adaptive water stress index is introduced to reflect interannual variations in water scarcity in future WSF estimations. In this framework, the WSI is no longer treated as a fixed parameter but evolves dynamically over time on the basis of projections of both future water use and water availability. Specifically, total water use is influenced by socioeconomic development and multisectoral water-saving policies, whereas total water availability is driven primarily by projected changes in surface runoff under climate change scenarios.

In our previous work, we projected future WSI at the prefecture level across China. The water demand projections incorporate officially committed water conservation policies and anticipate shifts in economic structure [45]. For water availability, we used runoff projections from six CMIP6 climate models—CanESM5, FGOALS-g3, GFDL-CM4, IPSL-CM6A-LR, MPI-ESM1–2-HR, and MRI-ESM2–0—which were downscaled to the prefecture level via the variable infiltration capacity (VIC) hydrological model. To further account for total water resources, we estimated groundwater availability for 2020–2050 on the basis of surface water projections and surface-to-groundwater ratios published in China's Water Resources Bulletin [46].

The adaptive WSI framework captures spatial and temporal variations in water scarcity and enables a more realistic estimation of future water scarcity impacts. The projections align with the SSPs, including SSP1, SSP2, and SSP5, thereby reflecting a range of plausible climate and policy futures.

2.5. Scenario sets for future water footprint analysis

To further investigate the future changes of Chinese data centers' WFs and the impact of the EDWC strategy on water scarcity, we constructed three scenario sets, which follow the development trajectories of the Baseline, Promotion, and Ambition respectively. Each scenario set consists of seven scenarios. We have 7 × 3 = 21 scenarios. Table 1 presents the parameter settings for the seven scenarios under the baseline trajectory. We focus on PUE improvements, WUE improvements, low-carbon electricity transition (LCT), and computational power distribution adjustments under the EDWC strategy. A more detailed breakdown of the parameters for all 21 scenarios can be found in Table S7 of the supplementary information.

In the PUE scenario, the national average PUE was 1.51 in 2022. We use a logistic function to reflect the different phases of technological progress, which captures a more realistic S-curve pattern of technological adoption: an initial slow decline, a rapid improvement phase as a new technology scale, and a stabilization period as the efficiency gains plateau. We estimate regional PUE values for six major regions (North, East, South, Central, West, and Northeast China) at five-year intervals from 2025 to 2045, as shown in Fig. S2. According to the Action Plan in the Information and Communications Industry pronounced by Chinese

Table 1
Key parameter settings for the core scenarios.

	ID	PUE	WUE	Electricity structure	Regional structure of newly added computing power
Baseness as usual (BAU)	S1	Same as 2022	Same as 2022c	Same as 2022c	Same as 2022
Improving Power Usage Efficiency (PUE)	S2	Decreases non-linearly from 1.5 in 2022 to around 1.2 by 2050	Same as 2022	Same as 2022	Same as 2022
Improving Water Use efficiency (WUE)	S3	Same as 2022	Decreases non-linearly from 1.8 m^3/MWH in 2022 to around 0.9 m^3/MWH by 2050	Same as 2022	Same as 2022
Low carbon electricity transition (LCT)	S4	Same as 2022	Same as 2022	Based on future electricity structure transitions (Table S8-S13)	Same as 2022
Eastern Data and Western Computing (EDWC)	S5	Same as 2022	Same as 2022	Same as 2022	The eight national computing hubs account for 68 % of the total newly added computing power in 2050.
Improving Power Usage Efficiency + Improving Water use efficiency + Low carbon electricity transition (TEI)	S6	Decreases non-linearly from 1.5 in 2022 to around 1.2 by 2050	Decreases non-linearly from 1.8 m^3/MWH in 2022 to around 0.9 m^3/MWH by 2050	Based on future electricity structure transitions (Table S8-S13)	Same as 2022
All measures above (COM)	S7	Decreases non-linearly from 1.5 in 2022 to 1.15 in the eight national computing hubs and 1.2 in other regions by 2050.	Decreases non-linearly from 1.8 m^3/MWH in 2022 to around 0.9 m^3/MWH by 2050	Based on future electricity structure transitions (Tables S7-S12)	The eight national computing hubs account for 68 % of the total newly added computing power in 2050.

government, the national average PUE is projected to decline to 1.1–1.2 by 2050 [31]. In the WUE scenario, we refer to the national standard, Data Center Resource Utilization Part 6: Water Resource Use Efficiency, to set the onsite water use efficiency for cooling [32]. In the Baseline, Promotion, and Ambition scenario sets, by 2050, the onsite WUE for the cooling system will reach 0.9, 0.85 and 0.8 m^3/MWH, respectively. Similarly, as shown in Fig. S3, we use a logistic function to interpolate the values at 5-year intervals from 2025 to 2045. To account for uncertainty, we conduct a sensitivity analysis on the k parameter, which controls the rate of technological advancement, setting its range between 1.5 and 2.5, based on historical efficiency improvement trends (Fig. S5).

In the LCT scenario, offsite water use efficiency in electricity generation is determined by the power mix. Renewable energy such as wind and solar power consume significantly less water per unit of electricity than nonrenewable energy such as thermal power. As China's power sector shifts from coal to renewables, offsite water intensity is expected to decline. We adopt the projected electricity mixes that simulate the regional power grid decarbonization process [47]. These projections align with national and provincial strategies and consider key factors such as renewable energy potential, types of thermal and hydropower generation, and interregional electricity transmission. Fig. S4 and Tables S7–S12 present the installed capacities of all power generation technologies by province from 2020 to 2050.

In the EDWC scenario, we incorporate the spatial shift in computing infrastructure outlined in the Chinese government's Eastern Data and Western Computing Project. This national strategy aims to alleviate energy and land constraints in eastern provinces by relocating computing resources to less developed, resource-rich western and northern regions. Specifically, eight national computing hub nodes have been designated: Beijing–Tianjin–Hebei, the Yangtze River Delta, the Greater Bay Area, Chengdu–Chongqing, Inner Mongolia, Guizhou, Gansu, and Ningxia [48]. These regions are expected to host over 60 % of the country's newly added computing power by 2025, rising to an estimated 70 % by 2050 [49]. While this redistribution may ease energy pressures in eastern urban centers, it is likely to increase water consumption in several water-scarce western regions. We define three scenarios—Baseline, Promotion, and Ambition—in which the eight hub regions account for 68 %, 70 %, and 72 % of the national new computing capacity by 2050, respectively (Table S7). This scenario design allows us to explore how large-scale digital infrastructure deployment under the EDWC initiative may reshape regional water demand and intensify water stress in ecologically fragile areas.

3. Results

3.1. Water footprints of Chinese data centers

In 2022, the total WFs of Chinese data centers was 15.7 billion m^3, accounting for 2.7 % of China's total water consumption. As shown in Fig. 2a, the provinces with the highest WFs were Hebei, Heilongjiang, Gansu, Inner Mongolia, and Sichuan, which contributed 11.2 %, 9.8 %, 7.8 %, 7.5 %, and 7.1 %, respectively, of the total national WFs for data centers. The prefectures with the highest WFs were Langfang, Harbin, Hohhot, Tianjin, and Lanzhou. Currently, the top five data centers with the largest WFs in the country are located in these cities. The total WF is influenced by two factors: first, the scale of computational power in the data centers. The computational power of these five prefectures accounts for 9.5 %, 8.2 %, 7.3 %, 6.5 %, and 7.8 %, respectively, of the total national computational power. Second, the WF intensity of the data centers, which is defined as the WF per unit of computational load. As illustrated in Fig. 2b, at the national scale, the average WF intensity of data centers is 0.039 m^3/watt (m^3/W). Prefectures in Northeast China and East China present relatively high water consumption intensities, at 0.048 m^3/W and 0.049 m^3/W, respectively, whereas those in Central China and South China present relatively low intensities, at 0.035 m^3/W and 0.037 m^3/W, respectively. The five prefectures with the highest water consumption were Langfang, Harbin, Tianjin, Hohhot and Lanzhou.

The WF of data centers comprises three components. First, the onsite WFs directly consumed during data center operation, such as cooling, were 10.6 billion m^3, accounting for 67.2 % of the total. Second, the offsite WFs indirectly consumed during electricity generation for data

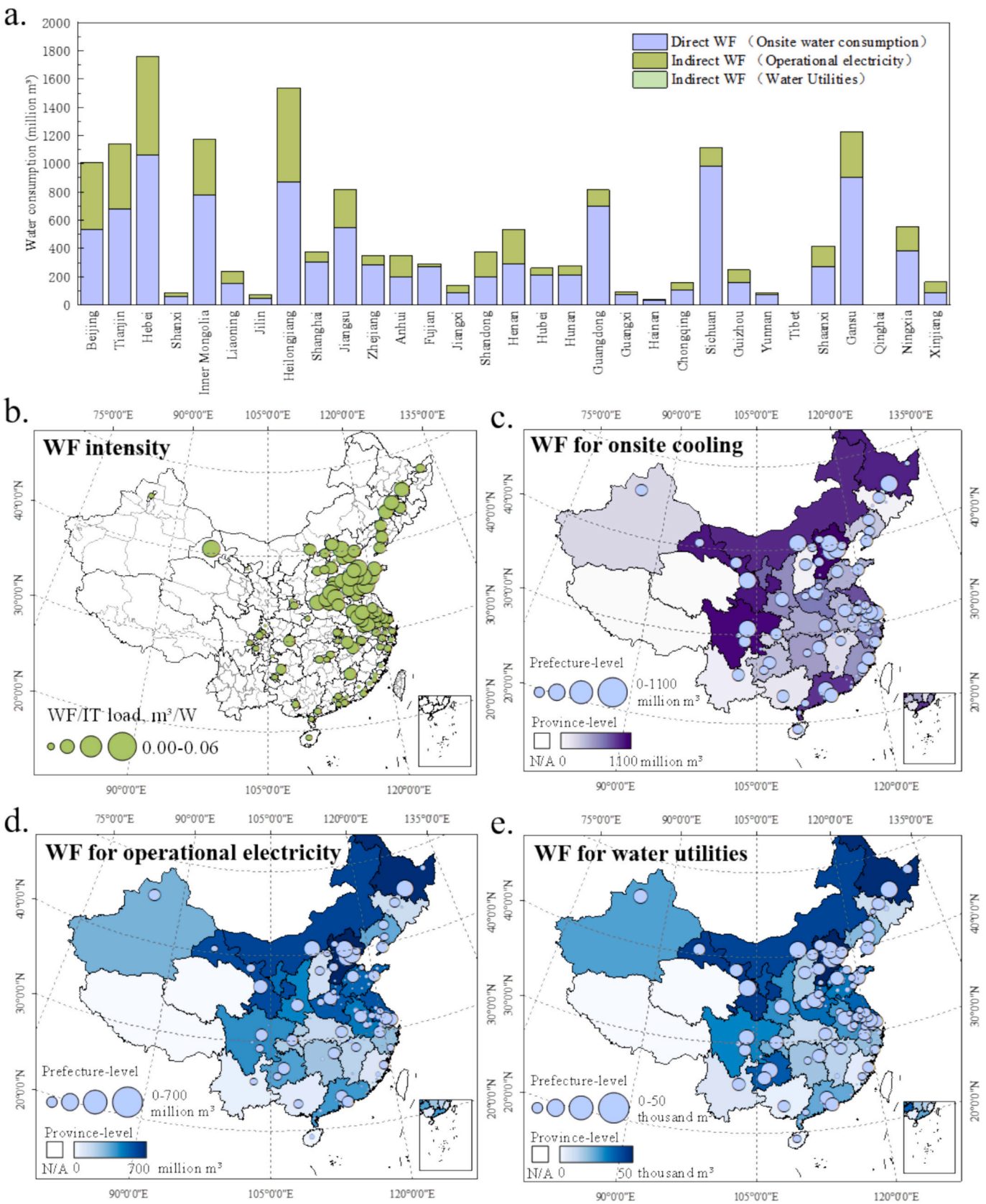


Fig. 2. The spatial distributions of the WFs and WF intensities for Chinese data centers. **a**. Province-level WFs and their composition. **b**. Prefecture-level WF intensity. **c**. Province-level and prefecture-level direct WFs. **d**. Province-level and prefecture-level indirect WFs for electricity generation used in data center operations. **e**. Province-level and prefecture-level indirect WFs for electricity generation used by water utilities.

center operation were 5.2 billion m^3, accounting for 32.8 % of the total. Third, the offsite WFs indirectly consumed during electricity generation for water and wastewater treatment utilities were 299.8 thousand m^3, making up only 0.002 % of the total. The first component is termed the direct WFs, whereas the latter two are classified as the indirect WFs. Fig. 2cde further illustrates the provincial- and prefecture-level distributions of these three types of WFs. The spatial distributions of the three WFs exhibit significant heterogeneity. The provinces with the highest onsite WFs included Hebei, Sichuan, and Gansu, each surpassing 900 million m^3. The provinces with the highest offsite WFs indirectly consumed during electricity generation for operation were Hebei, Heilongjiang, Beijing, and Tianjin, all of which exceed 450 million m^3. The provinces with the highest offsite WFs indirectly consumed during electricity generation for water utilities are Hebei, Heilongjiang, and Beijing, each surpassing 31.5 thousand m^3.

There are significant differences in the composition of WFs across provinces. In Heilongjiang, Jilin, Beijing, Tianjin, Henan, and Shandong, the proportion of direct WFs is very close to that of indirect WFs, with average distribution ratios of 55.9 % and 44.1 %, respectively. These regions rely primarily on thermal power generation, which entails a significant demand for cooling water during the electricity generation process, resulting in a relatively high proportion of indirect WFs. In Guangdong, Guangxi, Hainan, Sichuan, and Yunnan, the proportion of direct WFs far exceeds that of indirect WFs, with average distribution percentages of 86.5 % and 13.5 %, respectively. For example, in 2022, Fuzhou has the direct WFs of 194.4 million m^3 and the indirect WFs of 17.6 million m^3. This is partly due to the relatively developed service sector economy in South China, leading to a greater demand for computational power. Additionally, these regions have relatively cleaner electricity structures, with a smaller proportion of thermal power generation and a greater proportion of nuclear and hydroelectric power.

3.2. Dependence of Chinese data centers on scarce water supplies

In 2022, the WSFs of Chinese data centers were 13.4 billion m^3, accounting for 85 % of the total WFs. As illustrated in Fig. 3, the spatial distribution of Chinese data centers closely aligns with the spatial distribution of water stress. Approximately 71.9 % of computing power in China is located in prefectures experiencing severe water scarcity. Furthermore, 84.9 % of the computing power is situated in prefectures with severe or more extreme water scarcity. More specifically, the provinces with the highest WSF are Hebei, Gansu, Inner Mongolia, Tianjin, and Heilongjiang, whereas the prefectures with the highest WSF are Langfang, Hohhot, Tianjin, Lanzhou, and Harbin. These areas are also the primary regions experiencing water scarcity. The average WSI

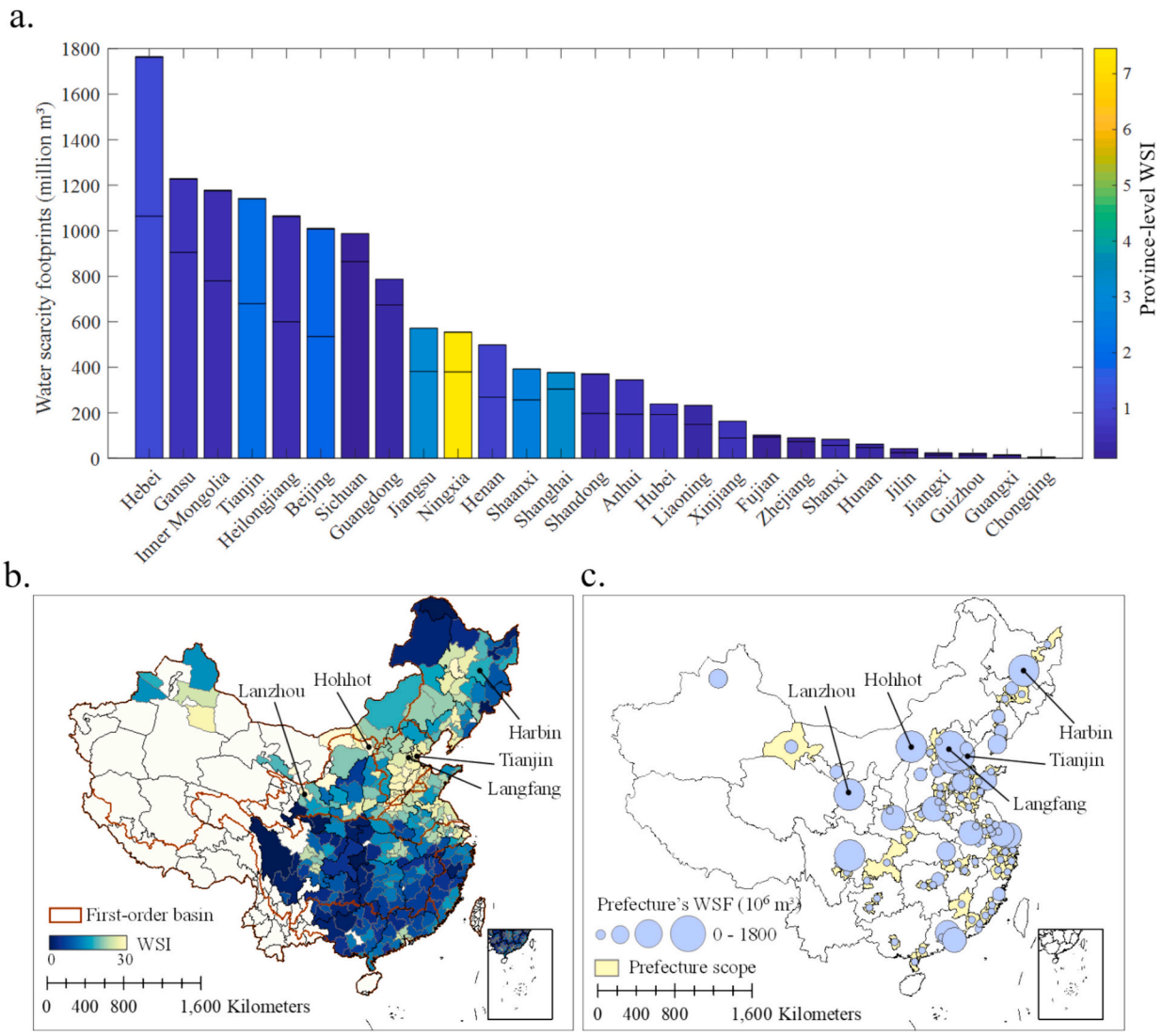


Fig. 3. The spatial distribution of the WSF for Chinese data centers. **a**. WSF and water scarcity of the province-level data centers. b. Prefecture-level water stress index. **c**. Prefecture-level WSF of data centers.

for China is 0.22, which is categorized as high water scarcity. The regions facing severe water scarcity are mainly concentrated in North China, Northeast China, and the eastern provinces of Jiangsu and Shandong. Among the 31 provinces, Ningxia, Shaanxi, Jiangsu, and Shanghai face the most significant water scarcity. Except for Shaanxi, the other three provinces fall under the severe water scarcity category, and they also exhibit large WSFs (360–600 million m^3). Fig. 3b and c show that the data centers in Shaanxi Province are located primarily in Xi'an, where the WSF of the data centers is 391.5 million m^3 and the WSI is 0.69, indicating significant water scarcity.

Nationally, the average WSF per unit of computing load in 2022 was 0.027 m^3/watt (m^3/W). This metric reflects the water consumption associated with computing loads and provides an important reference for understanding the impact of data centers on water resources. Fig. 4 illustrates the relationship between the average WSF per unit of computing load and the WSI at the prefecture level in China. The top ten prefectures with the largest IT load capacities are Langfang, Harbin, Chengdu, Lanzhou, Hohhot, Tianjin, Beijing, Zhongwei, Guangzhou, and Suzhou. Among them, nine, with the exception of Harbin, face severe water scarcity. More critically, these nine prefectures are designated new national computing hubs, indicating that they will bear even greater computing loads in the future, thereby exacerbating water resource tensions. Specifically, Beijing, Langfang, and Tianjin have the highest WSFs per unit of computing load, at 0.048, 0.042, and 0.042 m^3/W, respectively, significantly exceeding the national average. The WSFs for Lanzhou and Zhongwei are 0.034 and 0.037 m^3/W, respectively, which are slightly above the national average. These prefectures experienced severe water scarcity, with WSI of 4.5 and 5.0, indicating extremely severe water scarcity. As the Eastern Data and Western Computing Project advances, increasing computing capacity in these regions will undoubtedly further escalate water consumption, exacerbating the already strained water resource situation. Therefore, future predictive analyses of data center WFs and evaluations of related measures are crucial for the orderly operation of data centers and the management of local water resources.

3.3. Evolution of data center water footprints from 2025 to 2050

We quantify the long-term impacts of different strategies on Chinese data centers' WFs under 21 scenarios, derived from combinations of PUE improvements, WUE improvements, low carbon electricity transitions, and the EDWC strategy. The Baseline, Promotion, and Ambition trajectories represent varying levels of computing demand and intervention strength, thereby capturing uncertainties in future projections. As shown in Fig. 5a, the total WFs of China's data centers increase across all the scenarios, ranging from 32.1 to 40.0 billion m^3 in 2025 to 91.3–289.4 billion m^3 by 2050.

The BAU scenario highlights the unsustainability of current development trends. Without any intervention measures, the total WFs are projected to surge from 35.3 to 38.4 billion m^3 in 2025 to 192.9–272.1 billion m^3 in 2050, representing a 4.3- to 6.7-fold increase (Fig. 5a). This dramatic increase is driven by two main components: direct onsite water use for cooling accounts for 66.5 % of the total increase, whereas indirect offsite water use related to electricity use contributes to the remaining 33.5 %.

PUE improvements can significantly mitigate WF growth by reducing both onsite and offsite water use. When the national average PUE improves from a baseline of 1.8 in 2025 to 1.1–1.2 by 2050, the total WFs increase to 163.3–223.7 billion m^3, corresponding to a 3.5- to 5.4-fold increase. Compared with the BAU scenario, PUE optimization leads to a 10.5–17.8 % reduction in total WFs. A comparison of the Baseline (S2), Promotion (S9), and Ambition (S16) scenarios, featuring progressive PUE improvements to 1.2, 1.15, and 1.1, respectively, reveals diminishing marginal water-saving benefits. This finding underscores the limitations of relying solely on PUE improvements, which may be constrained by the trade-off between technological upgrade costs and achievable savings.

Improving WUE reduces water consumption by lowering direct

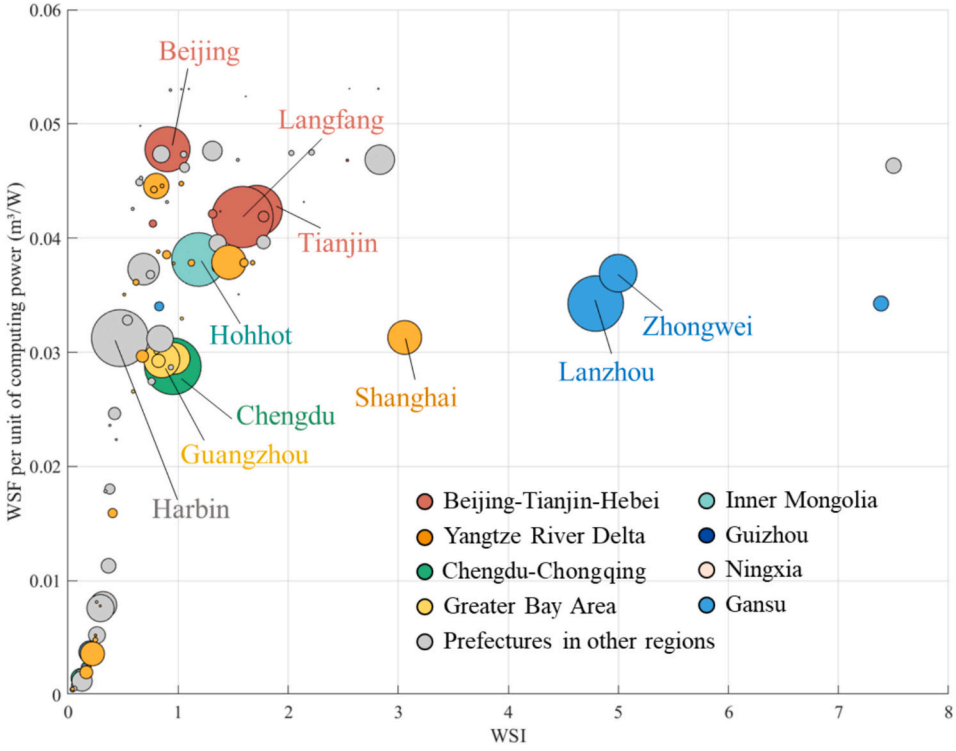


Fig. 4. Relationship between water scarcity footprint per unit of computing power and water stress index across Chinese data centers. Colored dots represent prefectures designated as new national computing hubs approved in 2022, while gray dots indicate other prefectures. The size of each dot corresponds to the IT load capacity of the respective prefecture.

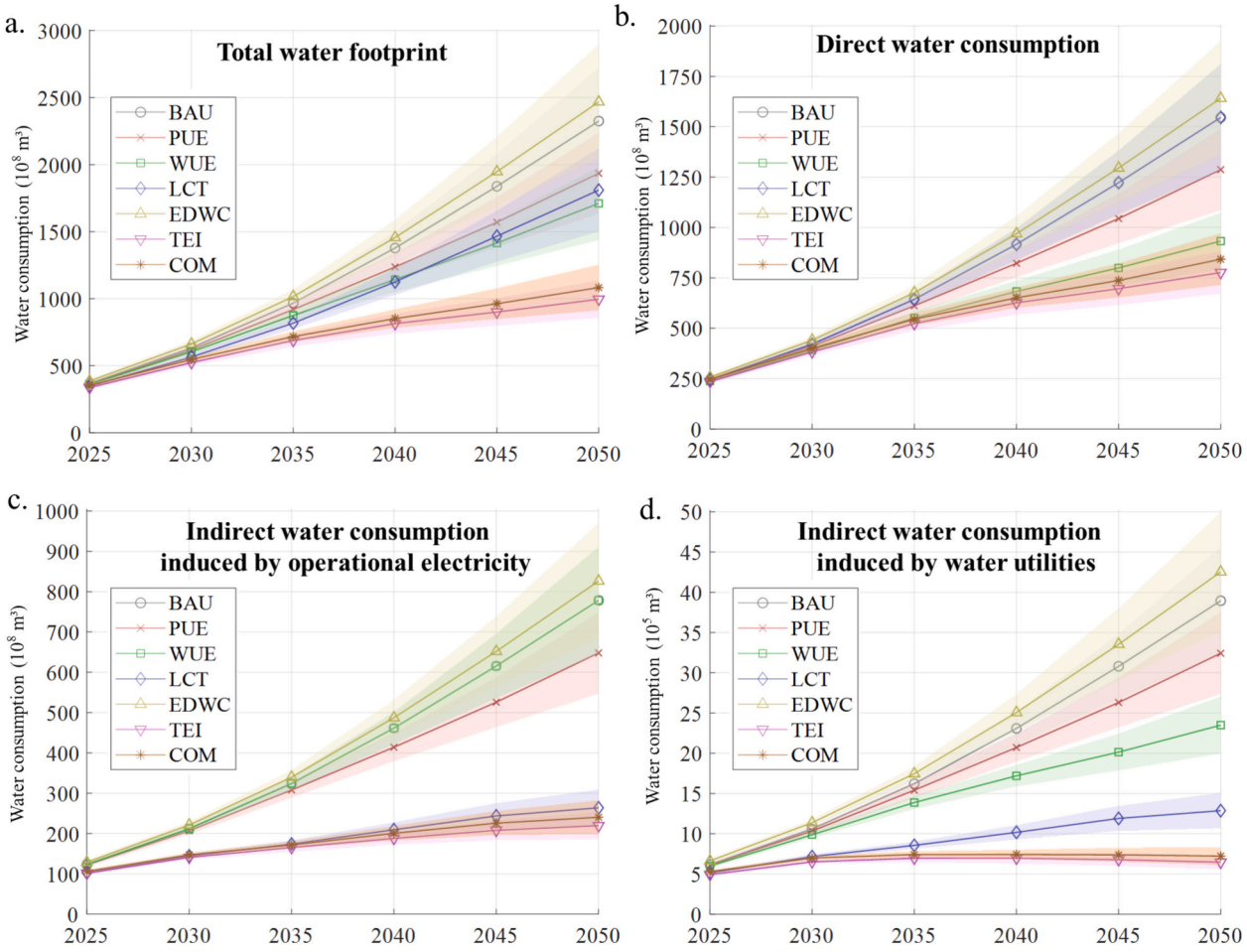


Fig. 5. National WFs of data centers in China from 2025 to 2050. **a**, Total WFs. **b**, Direct WFs. **c**, Indirect WFs induced by operational electricity generation. **d**, Indirect WFs induced by electricity use of water and wastewater treatment utilities. The different colored bands represent the range of WFs uncertainty under various scenarios derived from the baseline, promotion, and ambition scenario sets. The line represents the median value within this range.

onsite water use. Between 2025 and 2050, when the national average WUE decreases from 1.8 to 0.8–0.9 m^3/MWh, the total WFs of data centers increase from 34.6 to 37.7 billion m^3 in 2025 to 143.9–198.3 billion m^3 in 2050. Under the baseline, promotion, and ambition scenario sets, compared with the BAU scenario, WUE improvements reduce the total WFs by 23.7 %, 25.4 %, and 27.1 %, respectively. These reductions are attributed entirely to the decline in onsite WFs, corresponding to savings ranging from 49.0 to 72.7 billion m^3.

Decarbonization of the electricity sector contributes to water savings by increasing the share of low-water-intensity renewable energy in electricity generation. Between 2025 and 2050, the share of thermal power generation decreases from 56.0 % to 16.3 %, whereas the share of wind and solar power increases from 12.9 % and 11.6 % to 29.2 % and 40.8 %, respectively [47]. Compared with the BAU scenario, this energy transition reduces the total WFs of data centers by 18.2–22.1 %, driven entirely by reductions in the offsite WF, amounting to 42.7–60.3 billion m^3 of water savings.

The isolated implementation of computing power migration under the EDWC strategy may exacerbate regional water stress in China. In the EDWC scenarios, the total WFs of data centers increase from 37.1 to 40.0 billion m^3 in 2025 to 204.2–289.3 billion m^3 by 2050, exceeding the projected WFs under the BAU scenario. China's Eastern Data and Western Computing Project involves establishing computing hubs in eight regions: Beijing-Tianjin-Hebei, the Yangtze River Delta, the Guangdong-Hong Kong-Macao Greater Bay Area, Chengdu-Chongqing, Inner Mongolia, Guizhou, Gansu, and Ningxia. Under the baseline, promotion, and ambition trajectories, these regions are projected to host 68 %, 70 %, and 72 % of China's newly added computing capacity by 2050, respectively. Without improvements in PUE, WUE, or low-carbon electricity transition, the regional redistribution of computing capacity leads to 5.5 %, 5.9 %, and 6.3 % increases in total WFs compared with the BAU scenario. This increase is due primarily to higher PUE levels in the central and western regions than in the eastern provinces, resulting in elevated water consumption when computing workloads are shifted inland. More concerningly, several EDWC target provinces in western and northern China, such as Inner Mongolia, Gansu, and Ningxia, are already classified as water-scarce areas, with per capita water resources amounting to only one-fourth of the national average. The large-scale development of data center clusters in these regions creates a mismatch between high water demand and limited water availability, posing substantial challenges for local water supply security.

Combining improvements in PUE, WUE, and electricity decarbonization yields the greatest water savings. Among the seven scenario types compared, the TEI scenario results in the lowest total WFs for data centers, reaching 85.7–113.5 billion m^3 in 2050, a 51.5 %–58.3 % reduction compared with the BAU scenario. This scenario integrates three key mitigation measures: optimized PUE, enhanced WUE, and a low-water-intensity energy transition. Together, these measures target both direct and indirect components of the WFs. Direct WFs are reduced by 61.1–92.8 billion m^3 (accounting for 54.7 %–58.5 % of the total reduction), whereas indirect WFs decrease by 46.1–65.7 billion m^3 (41.5 %–45.3 %). When accounting for the regional redistribution of computing power under the EDWC strategy, the COM scenario results in total WFs of 91.3–125.3 billion m^3 in 2050, representing a 51.3 %–54.0

% reduction relative to BAU but a 0.4 %–10.3 % increase compared with the TEI scenario. These findings underscore that the implementation of EDWC must be accompanied by simultaneous improvements in PUE, WUE, and coal-to-renewable energy transition to effectively mitigate regional water stress.

3.4. Spatial distribution of water footprints for data centers in 2050

We assess the spatiotemporal evolution of facility-level WFs for data centers across 21 scenarios (Fig. 6, Fig. S6 and Fig. S7). The Jing-Jin-Ji region, Yangtze River Delta, and Pearl River Delta emerged as the core hotspots of WFs under all scenario sets, with consistently high WFs projected in Beijing, Hebei, Jiangsu, Shanghai, and Guangdong. Notably, the projected spatial pattern in 2050 diverges significantly from that in 2022. In 2022, data centers in Jiangsu and Shanghai contributed only 5.2 % and 2.4 % of the national total WFs, respectively. By 2050, these shares increase substantially under the three scenario sets, reaching 10.5–10.7 % for Jiangsu and 6.9–8.6 % for Shanghai. This sharp rise is driven primarily by the rapid economic expansion of the Yangtze River Delta, which fuels the growing demand for computing capacity. Located at the heart of this economic hub, Suzhou and Shanghai are expected to become major centers of WF growth due to their strategic advantages in data exchange and resource sharing. Under the COM scenario of the three sets, the WFs of Shanghai and Suzhou in 2050 are projected to reach 78.0–107.7 and 55.3–76.5 million m^3, respectively—second only to Beijing (126.1–176.2 million m^3) and Langfang (115.0–158.3 million m^3).

Efficiency-oriented strategies—including improvements in PUE, WUE, LCT, and the integrated TEI approach—can significantly reduce total WFs but are insufficient to alter the spatial concentration of high-consumption regions. Under the TEI scenario, all cities exhibit a WF reduction exceeding 40 % compared with the BAU scenario, resulting in nationwide water savings of 107.2–158.6 billion m^3 under baseline, promotion and ambition scenario sets. The greatest reduction is observed in Langfang, with a 61 % decline under ambition scenario set. As shown in Fig. 6a, Fig. S6 and Fig. S7, key data center hubs such as the Jing-Jin-Ji and Yangtze River Delta regions all continue to exhibit high WFs. This finding indicates that even with substantial technological optimization, regions with intense computing demand remain major resource consumers, underscoring the persistent spatial mismatch between digital infrastructure and environmental capacity.

The EDWC strategy, which reallocates computing capacity to the western and northern regions, leads to an overall increase in national WFs while simultaneously increasing local water demand. Among the 14

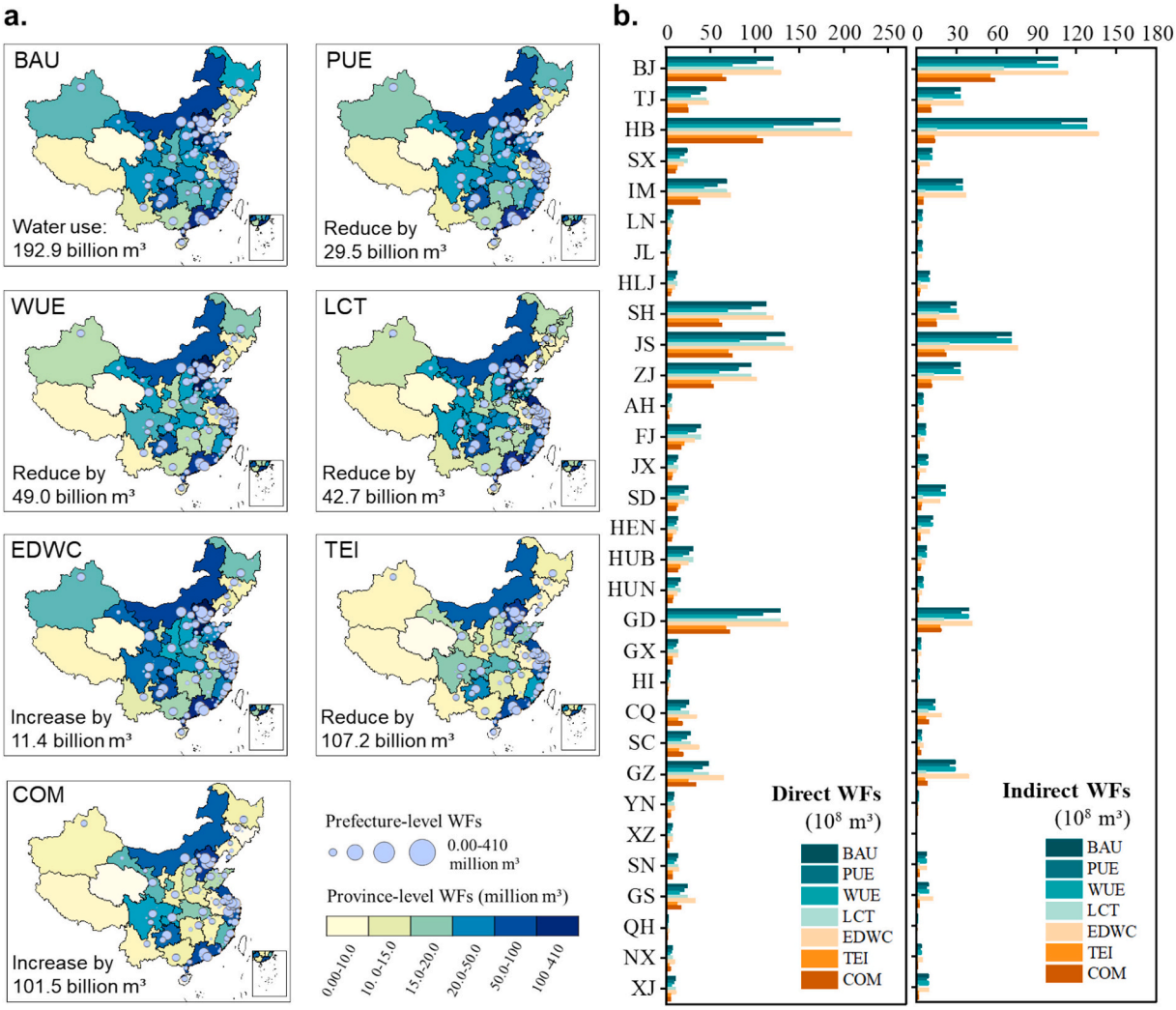


Fig. 6. Water footprints of data centers in China under seven scenarios of the baseline scenario set in 2050. a. Spatial distribution of WFs at the provincial and prefectural levels under seven scenarios. Background shading indicates provincial-level WFs, whereas circle size denotes prefectural-level WFs. The values represent changes relative to the BAU scenario. b. Composition of WFs, categorized into direct water consumption and indirect water consumption. The spatial distributions of WFs for the Promotion and Ambition scenario sets are shown in Supplementary Fig. S6 and Fig. S7.

provinces designated as new computing hubs under the EDWC strategy, the highest WFs are consistently observed in the EDWC scenario. Across the baseline, promotion, and ambition scenario sets, the EDWC strategy results in total WFs increases of 5.5 % (Fig. 6), 6.9 % (Fig. S6), and 8.3 % (Fig. S7), respectively, in major data center regions such as Jing-Jin-Ji, the Yangtze River Delta, Inner Mongolia, and Guangdong. In water-scarce provinces, including Guizhou, Sichuan, Chongqing, Gansu, and Ningxia, the corresponding increases reach 33.3 % (Fig. 6), 35.0 % (Fig. S6), and 36.8 % (Fig. S7), respectively. The largest increases are observed in Guiyang, Chongqing, and Chengdu, where WFs under the EDWC scenario reach 89.7–128.2, 52.9–75.7, and 36.1–51.6 million m^3, each exceeding the BAU levels by more than 30 %.

In contrast, the COM scenario, which integrates both efficiency improvements and spatial redistribution of computing workloads, achieves a nationwide WF reduction of 101.5–146.8 billion m^3. Compared with those of the baseline scenario set, the WFs of Guizhou, Sichuan, Chongqing, Gansu, and Ningxia are 45 %, 27 %, 30 %, 40 %, and 45 % lower, respectively. These results highlight that computing migration alone is insufficient to resolve water stress. The effectiveness of the EDWC strategy depends on the simultaneous optimization of the PUE and WUE, as well as the large-scale deployment of clean energy. Without these accompanying measures, the redistribution of computing workloads may not reduce the overall water footprint but instead exacerbate regional water stress.

3.5. Future water scarcity footprints for data centers

Under the BAU scenario of baseline, promotion and ambition sets, the WSF of data centers in China is projected to reach 191.8–270.6 billion m^3 by 2050 (Fig. 7a). When combined mitigation measures of PUE improvement, WUE improvement, and low-carbon electricity transition, are implemented (TEI scenario), the national WSF is significantly reduced to 85.1–112.8 billion m^3, a 52 %–58 % reduction relative to the BAU scenario of the three sets. When further incorporating the

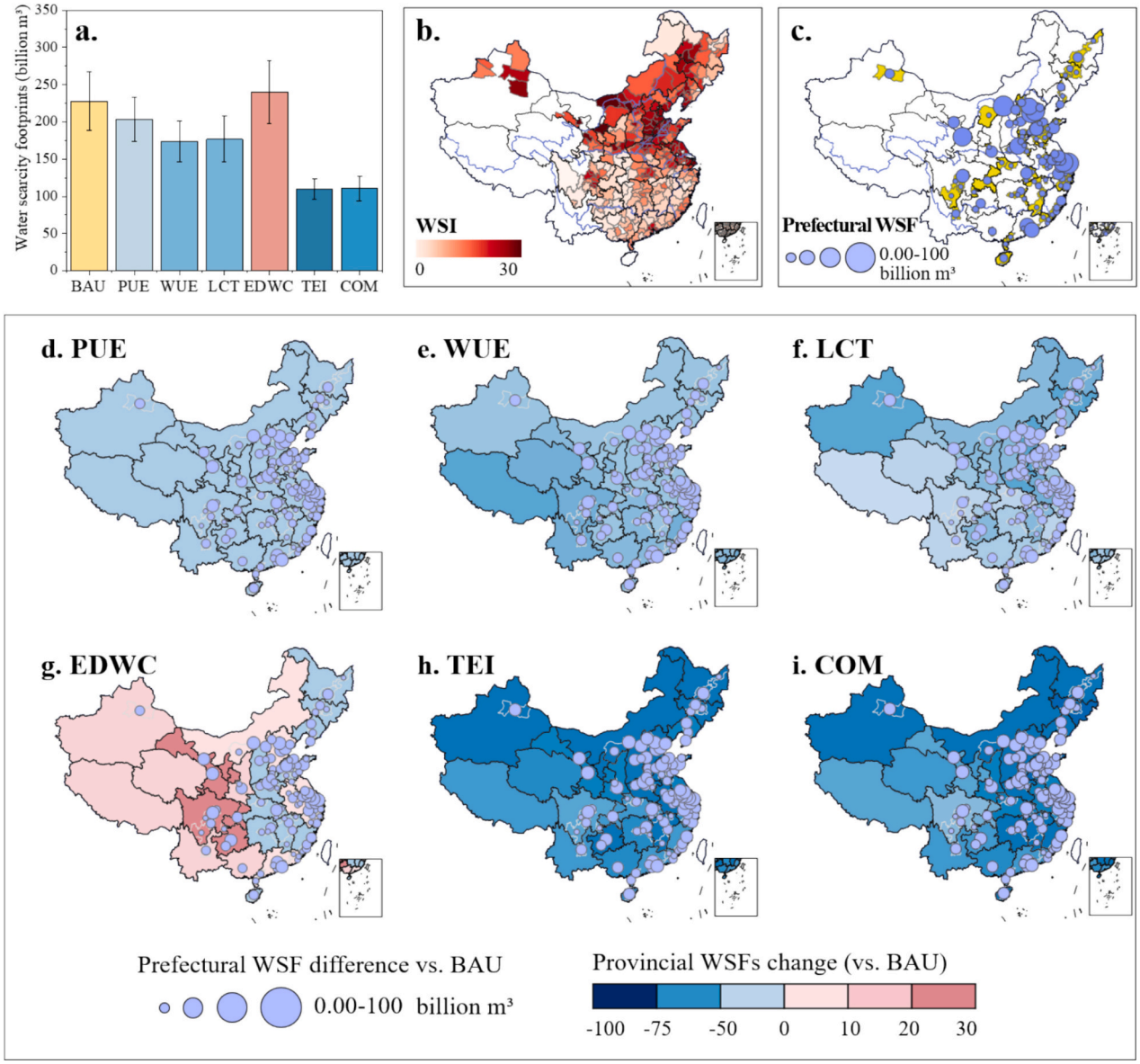


Fig. 7. Water scarce footprints of data centers in China under seven scenarios in 2050. a, Total WSFs under seven scenarios, with error bars indicating the uncertainty range across the baseline, promotion, and ambition scenario sets. b, Projected water stress index at the prefectural level in 2050. c, Prefectural-level WSFs under the BAU scenario along the baseline trajectory. d–i, Changes in WSFs under six mitigation scenarios (PUE, WUE, LCT, EDWC, TEI, and COM) relative to the BAU scenario, all under the baseline scenario sets. Background shading indicates provincial-level WSF changes, and circle sizes represent changes at the prefectural level.

effects of computing workload migration under the EDWC strategy (COM scenario), the WSF is slightly higher at 90.8–124.4 billion m^3 but still 51 %–54 % lower than the BAU of the three scenario sets.

The projected distribution of the water scarcity level in 2050 is shown in Fig. 7b. The national average WSI increases to 0.91, which is markedly higher than 0.78 in 2022. Nearly 70 % of cities experience worsening water scarcity, driven by reduced water availability due to climate change and increased demand from socioeconomic growth. We aggregate facility-level WSFs to the prefectural scale. As shown in Fig. 7c, the spatial distribution of WSFs under the BAU scenario reveals a degree of overlap between computing power hubs and water-scarce areas, particularly in North China and East China.

Figs. 7d–i illustrate the provincial and prefectural impacts of the six intervention scenarios relative to BAU. A cross-comparison of these panels highlights the most effective measures for each region. Overall, the TEI scenario achieves the greatest reduction in WSFs, with reductions exceeding 60 % in provinces such as Hebei, Shandong, Jilin, and Xinjiang. Among all the scenarios, only the EDWC strategy leads to WSF increases in several provinces—most notably in the west—due to the relocation of computing resources. In Chongqing, Sichuan, Guizhou, Gansu, and Ningxia, EDWC-driven workload migration results in WSF increases of over 30 % relative to BAU, underscoring the growing pressure on water-stressed regions.

The mitigation effectiveness of PUE improvement, WUE enhancement, and electricity decarbonization shows pronounced spatial heterogeneity. For example, reducing the onsite WUE to 0.9 m^3/MWh yields substantial water savings in Fujian (30.5 %), Qinghai (31.5 %), and Sichuan (31.6 %). In terms of the energy structure, following the national decarbonization roadmap, renewable energy is projected to reach 86.8 %, 82.1 %, and 99.0 % of total generation in Xinjiang, Shandong, and Hebei, respectively—resulting in WSF reductions of 37.6 %, 36.7 %, and 36.1 % by 2050, compared with BAU. These results highlight the need for region-specific mitigation pathways that combine technological upgrades with spatial planning to achieve effective and equitable reductions in data center-related water stress. It is important to note that the spatial distribution patterns of WSF under the Baseline, Promotion, and Ambition scenario sets are generally consistent. Therefore, this section focuses on discussing the general patterns observed across all three scenario sets.

4. Discussion

This study introduces a bottom-up framework to estimate the facility-level water footprints of data centers in China. We also evaluates the impact on regional water scarcity and forecasts future water footprints trends by considering evolving data center operations and shifting capacity across regions.This research fills a gap in Chinese data centers' water consumption accounting and highlights the necessity of incorporating water resource constraints in planning. This study estimates that in 2022, the total water footprints of data centers in China were 15.7 billion cubic meters, accounting for 2.7 % of the total water usage in China.

Our findings on the scale and composition of water footprints in Chinese data centers are generally consistent with prior estimates for other countries. In previous studies, Ristic et al. provided a rough calculation of the water footprint of data centers without considering spatial or operational unit differences and estimated that the annual water footprint of energy consumed by global data centers ranged from 767 million to 147,082 million cubic meters [25]. This range corresponds to the indirect WF calculated in this study. However, owing to the lack of technical detail in their approach, their results exhibit a high level of uncertainty and are not directly comparable. As there are currently no studies on the water usage of Chinese data centers, we compared our findings with those of Siddik et al. on the water footprint of U.S. data centers to validate the reasonableness of our results [9]. In 2018, the total water footprints of data centers in the United States were 1.29 billion cubic meters, accounting for approximately 3 % of national water usage, a proportion that is broadly consistent with the findings of this study.

Based on our findings, several measures can support more sustainable water management in data centers. First, the lack of detailed reporting highlights the need for mandatory disclosure of both direct and indirect water use, along with implemented water-saving strategies, to improve transparency [13]. Water footprint should be adopted as a key management metric and integrated into relevant policy frameworks. In 2018, China's data centers consumed 160.89 billion kWh of electricity, representing 2.35 % of national power use [50]. As Ristic et al. noted, indirect water use linked to electricity generation plays a critical role in overall water impacts [25]. Our facility-level analysis shows that direct cooling accounts for 10.6 billion cubic meters (67.2 %) of the total water footprint, while electricity-driven indirect use contributes 5.2 billion cubic meters (32.8 %). These results reinforce the need to include both direct and indirect water use when assessing environmental performance. The spatial variation in water footprints further reflects differences in local water-saving practices and power generation mix. Thus, any reporting requirement must consider both energy consumption and regional electricity profiles to avoid underestimation of impacts and unintended shifts in water stress across regions.

Second, local water availability and scarcity must be key considerations in data center siting. By overlaying spatial patterns of water scarcity and water footprints, we estimate that 13.4 billion cubic meters—85 % of total water use—occur in areas already facing water stress. Currently, 72 % of China's computing capacity is located in cities with severe water scarcity, and 85 % lies in regions with severe or extreme shortages. High and continuous cooling demand in provinces such as Hebei, Gansu, Inner Mongolia, Tianjin, and Heilongjiang further intensifies local water pressure, risking ecological degradation and reduced agricultural water supply. Similar trends have been observed in the U.S., where data centers disproportionately draw from water-stressed basins [9], underscoring the global nature of this challenge. To reduce risks, siting decisions should favor regions with more stable water resources—such as southwestern China (e.g., Sichuan, Yunnan, and Chongqing)—over drier regions like Gansu and Inner Mongolia [51]. Policymakers should encourage data center development in hydrologically resilient areas and promote water-saving technologies. Site planning should also consider colocation with renewable energy infrastructure to enable hybrid cooling systems and further reduce water demand [52,53].

Third, the integration of PUE optimization, WUE improvement, and grid decarbonization is essential for reducing the water footprints of data centers. Under the TEI scenario, their combined implementation reduces total water use by 51.5–58.3 % compared with the BAU scenario, significantly outperforming individual interventions. However, diminishing returns beyond a PUE of 1.15 highlight the limits of conventional cooling and the need for liquid cooling and heat recovery in high-impact regions such as Beijing and Hebei [54,55]. WUE optimization (from 1.8 to 0.8–0.9 m^3/MWh) enables scalable water savings, supporting the case of mandatory hybrid cooling and humidity-adaptive systems, especially in arid EDWC target regions such as Inner Mongolia and Ningxia [56]. Achieving 70 % renewable energy penetration reduces WFs by 18–22 %, reinforcing the need for policy frameworks that couple clean energy adoption with water efficiency. Promising alternatives such as underwater data centers, which utilize natural ocean cooling and eliminate active water use, may offer long-term benefits in coastal, water-stressed regions [57,58]. Provinces such as Tianjin, Guangdong, and Zhejiang, where high computing demand meets limited freshwater resources and land constraints, are well suited for piloting such solutions [59]. While still at an early stage, such technologies merit inclusion in future planning. Together, these strategies enable integrated water-energy management and support more sustainable digital infrastructure.

Finally, the effectiveness of the Eastern Data and Western Computing

Project (EDWC) in reducing the water footprint critically depends on concurrent improvements in PUE, WUE, and large-scale adoption of clean energy. Compared with the BAU scenarios, the separate implementation of computing load migration under the EDWC scenarios increased the total water footprint by 5.5–6.3 %, with water-scarce western regions such as Gansu and Ningxia experiencing water footprint surges exceeding 30 % due to their limited water resources. In contrast, the COM scenario—integrating EDWC with PUE/WUE optimization and electricity decarbonization—reduced national WFs by 51.3–54.0 % relative to BAU, demonstrating that spatial redistribution alone cannot resolve water conflicts without systemic efficiency upgrades. To mitigate regional water stress, policymakers must prioritize three measures. First, mandatory PUE and WUE thresholds should be enforced for new data centers in EDWC target regions, requiring the adoption of advanced cooling technologies and closed-loop water systems [60]. Second, computing capacity allocation should be linked to dynamic water scarcity assessments, prohibiting large-scale cluster deployments in areas with extreme water stress. Third, renewable energy adoption should be bound to EDWC implementation, ensuring that clean power procurement aligns with regional water conservation goals. These interventions could transform EDWC from a water-intensive risk into a sustainable pathway for digital infrastructure expansion.

5. Conclusion

This study presents the first bottom-up assessment of water footprints for Chinese data centers, capturing both direct onsite water use and indirect water use linked to electricity consumption. We map the geographic dependence of data centers on water-stressed basins, project water-footprint trajectories from 2025 to 2050, and evaluate how shifts in data center deployment may exacerbate regional water scarcity under future climate scenarios.

We found that China's data centers consumed 15.7 billion m^3 of water in 2022, with 72 % of computing capacity located in regions suffering from severe water scarcity. Under a business-as-usual trajectory, the Eastern Data and Western Computing Project would further increase water demand in arid provinces such as Gansu and Ningxia. However, by 2050, simultaneous improvements in power usage efficiency, water usage efficiency, and grid decarbonization could cut national water footprints by 51.5–58.3 %, offsetting the additional pressure from large-scale computing migration.

Although this study focuses on China, the bottom-up framework and scenario-based evaluation approach developed here can be applied to other regions. Future studies can adopt our modelling structure in different geopolitical contexts, such as Western Europe or North America. Such cross-regional applications would allow the identification of globally transferable best practices and help inform international data center siting and water resource management strategies.

CRediT authorship contribution statement

Feng Jiang: Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Cuncun Duan:** Conceptualization. **Bin Chen:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2025.126522.

Data availability

Data will be made available on request.

References

[1] Statista. Data centers - statistics & facts. *https://www.statista.com/topics/6165/data-centers/*.
[2] Jin C, Bai X, Yang C, Mao W, Xu X. A review of power consumption models of servers in data centers. Appl Energy 2020;265:114806.
[3] COMARCH, Top Data Center Trends. Global Overview. *https://www.comarch.com/trade-and-services/ict/news/top-data-center-trends-2025-global-overview/*; 2025.
[4] Masanet E, Shehabi A, Lei N, Smith S, Koomey J. Recalibrating global data center energy-use estimates. Science 2020;367(6481):984–6.
[5] Reinsel D, Gantz J, Rydning J. The digitization of the world from edge to core. IDC white paper 2018;16:1–28.
[6] Jones N. How to stop data centres from gobbling up the world's electricity. Nature 2018;561(7722):163–6.
[7] Van Heddeghem W, Lambert S, Lannoo B, Colle D, Pickavet M, Demeester P. Trends in worldwide ICT electricity consumption from 2007 to 2012. Comput Commun 2014;50:64–76.
[8] Belkhir L, Elmeligi A. Assessing ICT global emissions footprint: trends to 2040 & recommendations. J Clean Prod 2018;177:448–63.
[9] Siddik MAB, Shehabi A, Marston L. The environmental footprint of data centers in the United States. Environ Res Lett 2021;16(6):064017.
[10] Dayarathna M, Wen Y, Fan R. Data center energy consumption modeling: a survey. IEEE Commun Surv Tutor 2016;18(1):732–94.
[11] Lohrmann A, Farfan J, Lohrmann C, Kölbel JF, Pettersson F. Troubled waters: estimating the role of the power sector in future water scarcity crises. Energy 2023;282:128820.
[12] Islam MA, Ahmed K, Ren S, Quan G. Exploiting Temporal Diversity of Water Efficiency to Make Data Center Less" Thirsty". In: 11th International Conference on Autonomic Computing (ICAC 14); 2014.
[13] Mytton D. Data Centre water consumption. Npj Clean Water 2021;4(1):11.
[14] Yu H-C, Kuo L, B L. Ma The drivers of corporate water disclosure in enhancing information transparency. Sustainability 2020;12. https://doi.org/10.3390/su12010385.
[15] China Academy of Information and Communications Technology. White Paper on China's Computing Power Development Index. 2023. p. 2023.
[16] Horizon, China Data Center Market Size & Outlook, 2024-2030. *https://www.grandviewresearch.com/horizon/outlook/data-center-market/china*; 2025.
[17] Wang D, Zhou T, Lan F, Wang M. ICT and socio-economic development: evidence from a spatial panel data analysis in China. Telecommun Policy 2021;45(7):102173.
[18] China Internet Association. The Report on the Development of the Internet in China. 2023. p. 2023.
[19] Zhang N, Duan H, Guan Y, Mao R, Song G, Yang J, et al. The "eastern data and Western computing" initiative in China contributes to its net-zero target. Engineering 2024. https://doi.org/10.1016/j.eng.2024.08.010.
[20] Ma T, Sun S, Fu G, Hall JW, Ni Y, He L, et al. Pollution exacerbates China's water scarcity and its regional inequality. Nat Commun 2020;11(1):650.
[21] Masanet E, Shehabi A, Koomey J. Characteristics of low-carbon data centres. Nat Clim Chang 2013;3(7):627–30.
[22] Zhou F, Wang R, Ma G. Carbon emission scenario analysis of data centers in China under the carbon neutrality target. Int J Refrig 2024;168:648–61.
[23] Vasques TL, Moura P, de Almeida A. A review on energy efficiency and demand response with focus on small and medium data centers. Energ Effic 2019;12:1399–428.
[24] Rong H, Zhang H, Xiao S, Li C, Hu C. Optimizing energy consumption for data centers. Renew Sust Energ Rev 2016;58:674–91.
[25] Ristic B, Madani K, Makuch Z. The water footprint of data centers. Sustainability 2015;7:11260–84. https://doi.org/10.3390/su70811260.
[26] Wang P, Cao Y, Ding Z. Resources Planning Strategies for Data Center Microgrid Considering Water Footprints. In: 2018 2nd IEEE Conference on Energy Internet and Energy System Integration (EI2); 2018.
[27] Depoorter V, Oró E, Salom J. The location as an energy efficiency and renewable energy supply measure for data centres in Europe. Appl Energy 2015;140:338–49.
[28] Cao Z, Zhou X, Hu H, Wang Z, Wen Y. Toward a systematic survey for carbon neutral data centers. IEEE Commun Surv Tutor 2022;24(2):895–936.
[29] hehabi A, Smith SJ, Masanet E, Koomey J. Data center growth in the United States: decoupling the demand for services from electricity use. Environ Res Lett 2018;13(12):124030.

[30] Liu Y, Wei X, Xiao J, Liu Z, Xu Y, Tian Y. Energy consumption and emission mitigation prediction based on data center traffic and PUE for global data centers Glob. Energy Interconnec 2020;3(3):272–82.
[31] Ministry of Industry and Information Technology of the People's Republic of China. The Action Plan for Green and Low-Carbon Development in the Information and Communications Industry (2022–2025). *https://www.gov.cn/zhengce/2022-08/26/content_5706915.htm*; 2022.
[32] The Chinese National Standard. Resource Utilization in Data Centers - Part 6: Water Usage Efficiency. *http://nits.org.cn/index/article/4117?type=hydt*; 2024.
[33] Ni W, Hu X, Du H, Kang Y, Ju Y, Wang Q. CO_2 emission-mitigation pathways for China's data centers. Resour Conserv Recycl 2024;202:107383.
[34] Koot M, Wijnhoven F. Usage impact on data center electricity needs: a system dynamic forecasting model. Appl Energy 2021;291:116798.
[35] Fisenko SP, Brin AA, Petruchik AI. Evaporative cooling of water in a mechanical draft cooling tower. Int J Heat Mass Transf 2004;47(1):165–77.
[36] Capozzoli A, Primiceri G. Cooling Systems in Data Centers: state of art and emerging technologies. Energy Procedia 2015;83:484–93.
[37] The Chinese National Standard. Maximum allowable values of energy efficiency and energy efficiency grades for data centers. *https://std.samr.gov.cn/gb/search/gbDetailed?id=CE1E6A1DD42158F6E05397BE0A0A68DF*; 2021.
[38] Siddik MAB, Chini CM, Marston L. Water and carbon footprints of electricity are sensitive to geographical attribution methods. Environ Sci Technol 2020;54(12):7533–41.
[39] Guo L, Huang B, Qiu J, Qiu S. Research on the characteristics of fresh water consumption in nuclear power plant. J Hydraul Eng 2013;5:615–21.
[40] Jin Y, Behrens P, Tukker A. Scherer L. Water use of electricity technologies: a global meta-analysis. Renew Sust Energ Rev 2019;115:109391.
[41] International Organization for Standardization (ISO) 14046 - Environmental management - Water footprint - Principles, requirements and guidelines. *https://www.iso.org/standard/43263.html*; 2014.
[42] Pfister S, Koehler A, Hellweg S. Assessing the environmental impacts of freshwater consumption in LCA. Environ Sci Technol 2009;43(11):4098–104.
[43] Scherer L, Pfister S. Dealing with uncertainty in water scarcity footprints. Environ Res Lett 2016;11(5):054008.
[44] Su B, Zhai J, Wang Y, Lin Q, Gao M, et al. Gridded value-added of primary, secondary and tertiary industries in China under shard socioeconomic pathways. Sci Data 2022;9(1):309.
[45] Jiang F, Chen B, Wang H, Duan CC. Mitigating China's prefecture-level economic risk of water scarcity: the role of water conservation and carbon neutrality policies. Resour Conserv Recycl 2025;215:108140.
[46] Ministry of Water Resources of P.R.C., China Water Resources Bulletin. Beijing: China: Water Power Press; 2022.
[47] Chen R, Xu P, Yao H. Decarbonization of China's regional power grid by 2050 in the government development planning scenario. Environ Impact Assess Rev 2023;101:107129.
[48] State Council of the People's Republic of China. Official Launch! Comprehensive Implementation of the "Eastern Data and Western Computing" Project. https://www.gov.cn/xinwen/2022-02/17/content_5674322.htmOfficial; 2022.
[49] National Development and Reform Commission. Implementation Opinions on Deeply Implementing the "Eastern Data and Western Computing" Project and Accelerating the Construction of a National Integrated Computing Power Network. *https://www.gov.cn/zhengce/zhengceku/202401/content_6924596.htm*; 2022.
[50] Greenpeace, Energy Consumption and Renewable Energy Potential of Data Centers in China. *https://www.greenpeace.org.cn/*; 2025.
[51] Zhao F, Wang X, Wu Y, Singh SK. Prefectures vulnerable to water scarcity are not evenly distributed across China Communic. Earth Environ 2023;4(1):145.
[52] Konapala G, Mishra AK, Wada Y, Mann ME. Climate change will affect global water availability through compounding changes in seasonal precipitation and evaporation. Nat Commun 2020;11(1):3044.
[53] Cui X, Yang C, Yan W, Wang X, Liu Y, Jin L. Climatic applicability of indirect evaporative cooling strategies for data centers in China. J. Build. Eng. 2024;83:108431.
[54] Ebrahimi K, Jones GF, Fleischer AS. A review of data center cooling technology, operating conditions and the corresponding low-grade waste heat recovery opportunities. Renew Sust Energ Rev 2014;31:622–38.
[55] Lu T, Lü X, Välisuo P, Zhang Q, Clements-Croome D. Innovative approaches for deep decarbonization of data centers and building space heating networks: modeling and comparison of novel waste heat recovery systems for liquid cooling systems. Appl Energy 2024;357:122473.
[56] Ndukaife TA, Nnanna AGA. Optimization of water consumption in hybrid evaporative cooling air conditioning Systems for Data Center Cooling Applications. Heat Transfer Engineering 2019;40(7):559–73.
[57] Periola AA, Alonge AA, Ogudo KA. Heat wave resilient systems architecture for underwater data centers. Sci Rep 2022;12(1):17161.
[58] Hu Z-H, Zheng Y-X, Wang Y-G. Packing computing servers into the vessel of an underwater data center considering cooling efficiency. Appl Energy 2022;314:118986.
[59] Wang Z, Ye J. *Power optimization system of underwater data center*. In *2023 international conference on network, multimedia and information technology (NMITCON)*. IEEE 2023:1–8.
[60] Kseibati R. Cooling innovation and circularity: Addressing water stress in the age of AI-driven data centers. Massachusetts Institute of Technology; 2025.